Contribution Journal 12

L03_Team8_ITAI1371 – Machine Learning Lab: Model Evaluation with Logistic Regression on the Titanic Dataset

Course: ITAI1371

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Overview

For this lab, our group built and evaluated a logistic regression model using the Titanic dataset. The focus was on model evaluation—the confusion matrix, classification metrics (accuracy, precision, recall, and F1-score), and cross-validation. Since this was a group project, we divided the deliverables so that each member owned one major piece of the work. Below, I describe in my own words exactly what each team member contributed.

Katherine Delores – Contribution Journal

My responsibility was the contribution journal—this document. I tracked how the work was divided from the start and recorded what each team member did so the final submission accurately reflects everyone's effort. I kept notes on our planning discussions, including why we agreed on logistic regression as our baseline model and why we chose k-fold cross-validation over relying on a single train/test split. I also coordinated with Marisabel to understand her workflow so my descriptions of the coding portion are accurate, and with Joshua to make sure his reflective journal aligned with the technical work actually completed. Finally, I reviewed all three deliverables—the notebook PDF, the reflective journal, and this contribution journal—to confirm they were properly named, clearly formatted, and consistent with each other. Writing this journal taught me that documentation is not just paperwork; it forces the team to be accountable and makes the project reproducible.

Marisabel Morales – Coding

Marisabel owned the coding portion of the lab and completed it independently from start to finish. She loaded the Titanic dataset and performed all the preprocessing: identifying and handling missing values, encoding

categorical variables, and selecting the features used in the model. She then split the data into training and test sets, built the logistic regression model with scikit-learn, and fit it to the training data. After that, she generated the confusion matrix and calculated the classification metrics—accuracy, precision, recall, and F1-score—so we could evaluate the model beyond simple accuracy. She also implemented cross-validation to get a more reliable estimate of performance and compared those results with the single-split evaluation. Throughout the notebook, she added comments explaining each step and produced a visualization of the confusion matrix, which made the results much easier for the rest of the team to interpret. Her clean, well-documented workflow was the backbone of the entire project.

Joshua Balderas – Reflective Journal

Joshua was responsible for the reflective journal, where he documented the team's learning process and the complexity of the workflow. He wrote about what went smoothly—such as implementing cross-validation and seeing how the metrics gave a fuller picture of model performance than accuracy alone—and what was challenging, including deciding how to handle missing data and understanding why precision and recall can tell different stories about the same model. He reflected on how the confusion matrix helped us see the specific types of errors the model made, and he connected the lab back to the learning types discussed in class and to the ML tools from the previous lab. He also noted possible next steps, like hyperparameter tuning or comparing against other models, and explained why certain choices were made in the workflow, such as why cross-validation is more trustworthy than a single split. His reflections tied the technical work back to the course's learning objectives and gave the project a thoughtful conclusion.

Summary

Our group divided the deliverables clearly: I handled the contribution journal, Marisabel built and evaluated the model, and Joshua wrote the reflective journal. Each person's work depended on the others—Marisabel's code and results gave Joshua the material to reflect on, and both gave me the information I needed to document everyone's contributions. This division kept us accountable and ensured the final submission covered the technical, reflective, and documentation sides of the lab.

This journal was written by Katherine Delores based on her own observations of the team's work.